

GRAND FINALE: TOPIC 60/60

The 60-Day GenAI Master Cheat Sheet.

Synthesizing everything from tokenization and prompt engineering to RAG pipelines and model finetuning into one definitive mental architecture.

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THE MACRO VIEW

The 4-Layer GenAI Stack



4. The Application Layer

Retrieval-Augmented Generation (RAG), Autonomous Agents, Vector Databases, Tool Integration (Calculators, Web Search).



3. The Steering Layer

Prompt Engineering, System Instructions, Few-Shot Demonstrations, Chain of Thought Reasoning, Persona Framing.



2. The Brain Layer

Foundation Models (GPT-4, Claude), Small Language Models (Llama 3 8B), Pre-training, LoRA Fine-Tuning.



1. The Data Layer

Tokenizers, Byte-Pair Encoding (BPE), Vocabularies, High-Dimensional Vector Embeddings.

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LAYER 1: THE INPUT

Data & Tokenization

Tokenization (BPE)

- **Models don't read words.** They read integers mapped to text chunks.
- **1 Token $\approx$ 0.75 Words.**
- **Byte-Pair Encoding:** Merges frequent characters into subwords (e.g., "ing", "tion").

Embeddings

- **Meaning as Math:** Words are converted into high-dimensional vectors (e.g., 1,536 dimensions).
- **Spatial Distance:** "King" and "Queen" live mathematically closer together than "King" and "Apple".



Critical Reality Check

Tokenizers are biased toward English. Non-English languages require significantly more tokens to represent the same concept, costing more money and eating up context windows faster.

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LAYER 2: THE ENGINE

The Model Spectrum

The core neural network that predicts the next token. You must choose between massive generalized intellect or compact, localized speed.

LLMs (70B+ Params)

GPT-4, Claude 3.5 Opus, Llama 3 70B

- ✓ Emergent multi-step reasoning.
- ✓ Massive global world knowledge.
- ✗ High latency & API pay-per-token cost.

SLMs (<10B Params)

Llama 3 8B, Phi-3 Mini, Qwen 2.5

- ✓ Edge-native (Runs on MacBooks/Phones).
- ✓ Near instant latency, Air-gapped privacy.
- ✗ High hallucination risk on complex logic.

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Prompting Fundamentals



THE DUAL-MESSAGE PARADIGM

1. System Prompt

The "Operating System". Establishes the persona, global rules, negative constraints, and output formatting. Defends against jailbreaks.

2. User Prompt

The specific task or data input. Wrapped in structural delimiters (like XML tags) to separate it from the system instructions.

Strategy	Definition	Best For
Zero-Shot	Direct instruction. No examples provided.	Translation, Summarization.
One-Shot	Instruction + 1 perfect example (Exemplar).	Forcing exact JSON formats.
Few-Shot	Instruction + 3 to 10 examples.	Complex styling, tone matching.



LAYER 3.5: ADVANCED CONTROL

Logic & Determinism



Chain of Thought (CoT)

Adding *"Let's think step by step"* forces the LLM to generate intermediate reasoning tokens, acting as a "scratchpad" memory.

Standard: Fails complex math.
CoT: Explains steps -> Succeeds.



Persistent Answers

LLMs are stochastic (random). To lock down persistent, deterministic outputs, you must adjust the API sampling parameters.

Temperature = 0: Greedy decoding (Most logical, zero creativity).

Top-P (Nucleus): Restricts vocabulary to only highly probable words.

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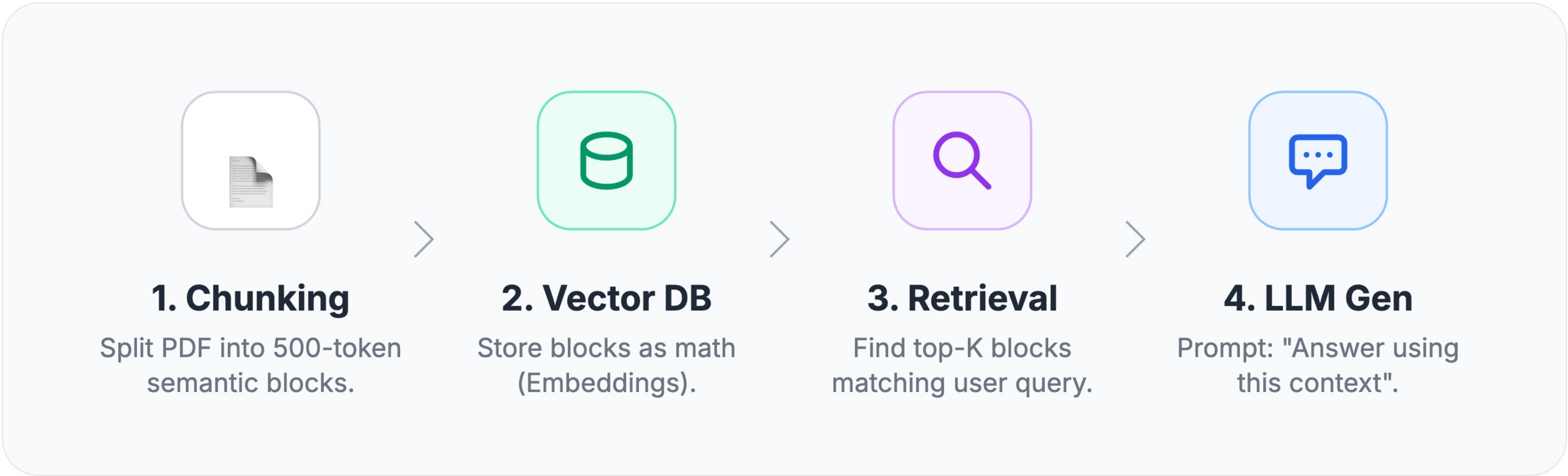
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LAYER 4: APPLICATION

The RAG Pipeline

Retrieval-Augmented Generation (RAG) acts as an open-book test for the LLM, giving it your private data right before it answers.



THE DECISION TREE

Model Customization Matrix

METHOD	MODIFIES MODEL?	PRIMARY GOAL	DATA REQUIREMENT
1. Prompting	No	Behavior & formatting steering.	0 to 10 examples
2. RAG	No	Injecting private/live facts.	Unstructured Docs
3. Fine-Tuning	Yes (Weights)	Domain mastery & tone lock.	100s - 10,000s Q&A pairs

Rule of Thumb: Try Prompting first. If it hallucinates facts, use RAG. If the tone/style is consistently wrong, Fine-Tune.



BEYOND CHATBOTS

The Agentic Future (ReAct)

The next evolution is moving from LLMs that just "talk", to Agents that Reason and Act.

The ReAct Loop (Reasoning + Acting)

THOUGHT

"I need to find the current stock price of Apple. I don't have real-time data, so I must use a tool."

ACTION

```
call_tool(name="web_search", query="AAPL stock price today")
```

OBSERVATION

"Search results returned: AAPL is currently trading at \$189.43."

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The End of the Beginning.

We've spent 60 days breaking down the AI ecosystem block by block. Now, you hold the blueprints. It's time to go build.

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